

Create automation script for Salesforce Create Lead functionality
Use Selenium Java with TestNG & Page Object Model.
Setup Pre-Condition and Post-Condition using TestNG annotations

Persona: Act as a Senior QA Automation Architect with over 12 years of experience in Selenium, Java, TestNG, Maven, and Salesforce automation. Follow enterprise automation framework standards, clean code principles, and industry best practices while explaining design decisions where necessary.

I – Instructions

Create a complete automation test script for the Salesforce Create Lead functionality using Selenium WebDriver with Java, TestNG, and the Page Object Model (POM) design pattern.

The solution should:

Follow Java coding standards and automation best practices.
Implement the Page Object Model with separate Page classes and Test classes.
Use explicit waits instead of static waits wherever applicable.
Include proper exception handling and logging.
Add meaningful comments throughout the code.
Make the framework modular, reusable, and maintainable.
Externalize test data wherever appropriate.

C – Context

The automation framework is being developed for a Salesforce application.

Technology Stack:

Java

Selenium WebDriver

TestNG

Maven

Page Object Model (POM)

Requirements

****[Mandatory]****Automate the Create Lead functionality.

Implement TestNG lifecycle annotations.

Use:[STRICT]

@BeforeSuite / @AfterSuite (if applicable)

@BeforeClass

@BeforeMethod

@AfterMethod

@AfterClass

****[Strict]****Setup all required preconditions before test execution.

Perform post-condition activities such as logout, browser closure, and resource cleanup.

[MANDATORY]Include assertions to validate successful lead creation.

Organize the project into standard automation framework packages such as:

pages

tests

utilities

base

constants

listeners (optional)

Example:

Launch the browser.

Navigate to the Salesforce login page.

Login with valid credentials.

Navigate to the Leads tab.

Click New Lead.

Enter mandatory Lead details:

First Name

Last Name

Company

Click Save.

Verify that the Lead is created successfully.

Logout from Salesforce.

Close the browser.

T – Tone

Use a formal, professional, and instructional tone suitable for enterprise software development and QA automation teams. Explain the purpose of each component clearly and provide production-ready, maintainable code following industry best practices.

O – Output Format

Provide the response in the following order:

Framework Architecture Overview

Project Folder Structure

Maven pom.xml

Base Test Class

Utility Classes

Page Object Classes

Test Class using TestNG

TestNG XML Configuration

Pre-Condition and Post-Condition Implementation

Assertions and Validation Logic

Execution Steps

Best Practices Followed

Possible Enhancements for Enterprise Frameworks